LiDAR Maze 10/14

Software + FPGA

Project overview

- This is the second quarter of this project
- We have some code + designs from last quarter
- More hands-on engagement this quarter
- We are building a small “car” (~10x10 inches) to autonomously solve a maze
- We must build the car and the maze
- Will be using github

Raise of hands: new to this project?

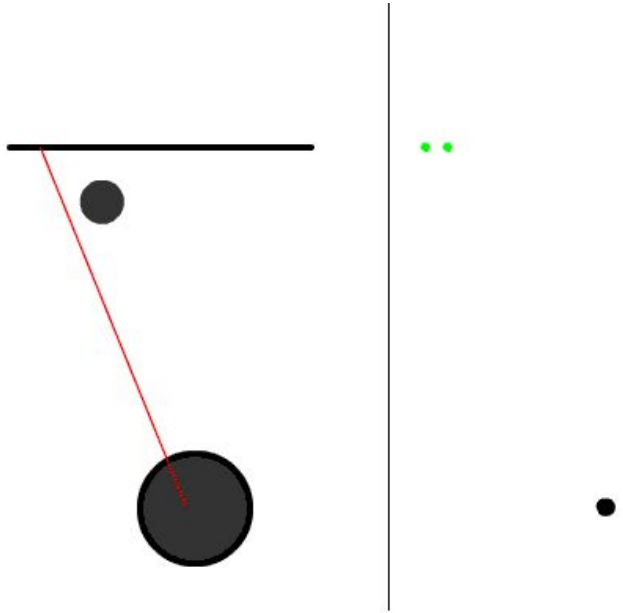


The roof-mounted unit installed on Waymo's ride-hailing vehicles contains various hardware components for autonomous driving. *Courtesy of Waymo*

Software subteam

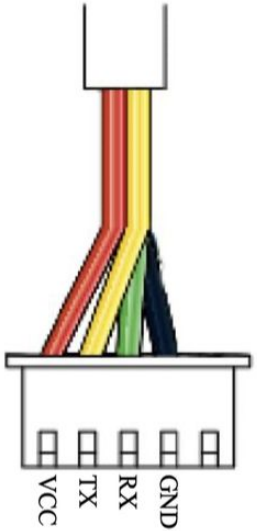
- Write maze-solving algorithms and test them, digitally and physically.
- Create a digital simulation of the car and maze environment to test different solving algorithms

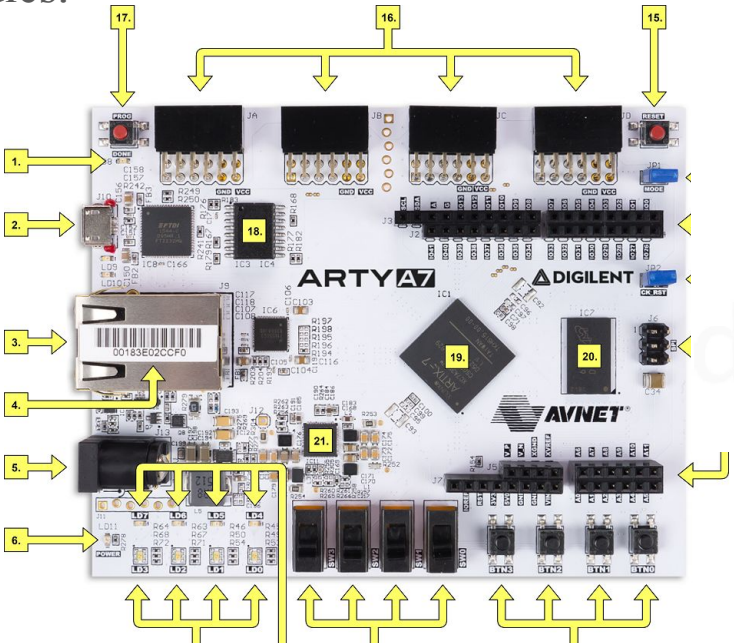
Leads: Akshay

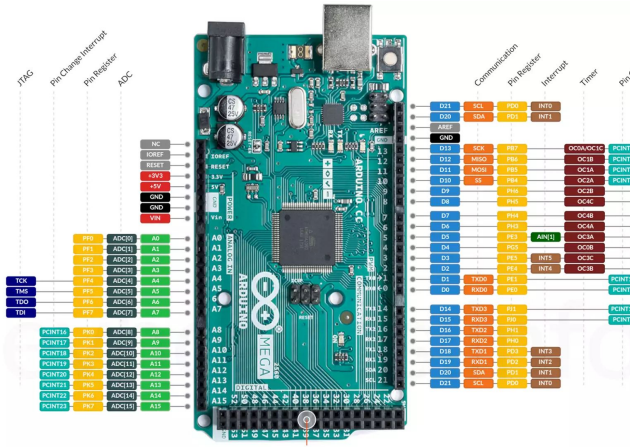


FPGA subteam

- Implement LiDAR -> FPGA -> Arduino data pipeline (including wiring)
- Write SystemVerilog code: LiDAR initialization, data packaging, and data transmission modules.







Leads: Alexa, Dara

Meeting today

- New members must complete a hands-on task to join their desired subteam
- Use the internet, AI, whatever you want to complete it
- PLEASE ask the leads questions (this isn’t a test! It shouldnt be silent in this room)
- This is to emulate how meetings will go. Leads are here to give you a task and help you, not be your tutor.
- You will have until the end of the meeting

task

Software: In groups of 3+: research and come up with a maze-solving algorithm that we could use. Imagine you are the robot solving a maze with your algorithm.

FPGA: Look up RP LiDAR C1, Arty A7 FPGA, Arduino Mega 2600 communication pinouts (datasheets). Determine how to wire them together.

Any questions? Ask your lead where to start!

(demo the lidar)

- Rn the lidar is connected to computer
- Eventual goal is lidar -> fpga -> arduino
- The fpga acts as a data repackager/processor
  - (enables parallel pathfinding and navigation on arduino)
  - Industry standard
  - yes
- Any questions?

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meeting wrap up

- Raise of hands: who would be interested in a soldering workshop?
- Any other questions? leads will stick around (introduce yourself)

attendance

Miss one: warning

Miss two: speak to lead

Miss three: kicked from project

Please communicate your absences! We are flexible

Code: wart

